



VC SpyGlass Project Import Kit Help Guide

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1 Introduction

This import kit provides automated infrastructure to help SpyGlass customers import their existing projects into VC SpyGlass. This is meant to help accelerate the customer setup and provide them with an easy way to start working with the next generation VC SpyGlass Static RTL Signoff Platform. In addition to the import function, this kit will provide the ability to see the SpyGlass and VC SpyGlass QoR differences on any design.

1.1 Starting Pre-requisites

Customers will need a design to be considered for import to the VC SpyGlass platform. The SpyGlass project under consideration should be using version O-2018.09 and above version.

- 1) Must provide top module name within the SpyGlass project file
- 2) If SpyGlass setup reads in .lib cells
 - a. Provide equivalent .db cells for import
 - b. Provide .db for VC SpyGlass, user needs to specify .synopsys_vcst.setup in design directory

Syntax for .synopsys_vcst.setup file

```
set search_path "< Path to directory which contain db files">
set link_library "<DB file list which located in search path>"
```

Example Syntax of .synopsys_vcst.setup

```
set search_path "<dir-path>/LIBRARIES_db/"
set link_library "LIB1.db LIB2.db"
```

Note: If you do not specify the .db file corresponding to the .lib files specified in the SpyGlass setup, VC SpyGlass will consider the lib cells as a black-boxes.

For more information see SolvNet article 2594331, "[Translating .lib file to .db files and using it in SpyGlass](#)"

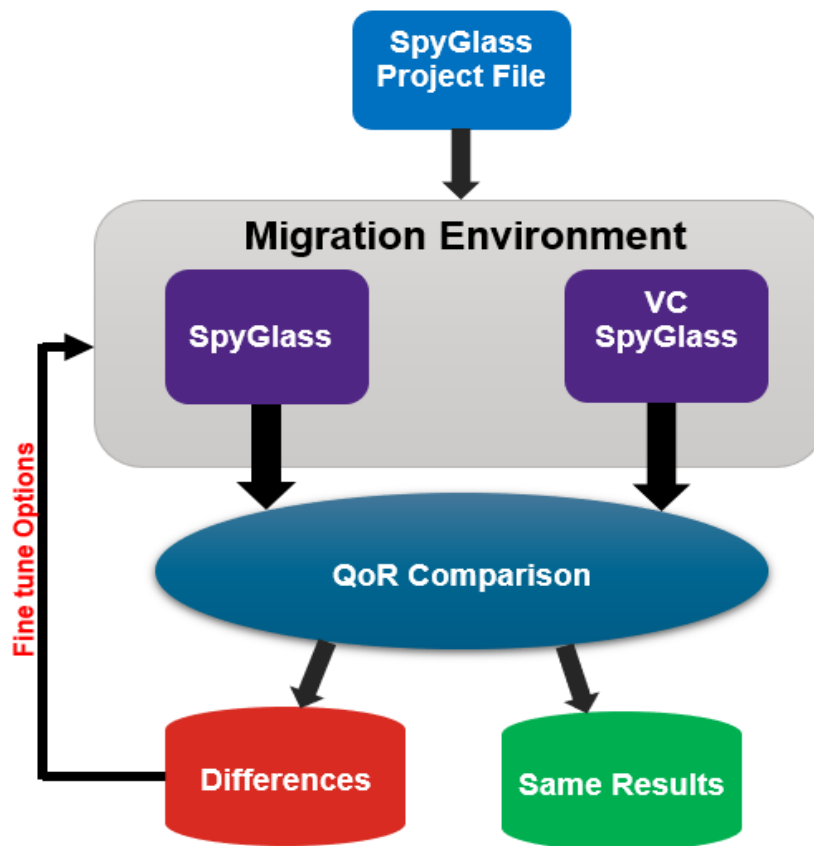
1.2 Setting Up Your Environment

Set the below environment variable on shell:-

```
%> setenv VC_SPYGLASS_PROJECT_IMPORT_KIT_DIR vcstatic_<release-  
name>/auxx/monet/scripts/vc_spyglass_project_import_kit  
  
>% setenv PATH $VC_SPYGLASS_PROJECT_IMPORT_KIT_DIR/scripts:$PATH
```

2 Migration Methodology

This import kit takes SpyGlass project file as input and gives the QoR comparison of desired products.



3 Comparison of Lint Results

Provide the SpyGlass project file as input to “vc_spyglass_import”, it creates SpyGlass and VC SpyGlass setup files to perform desired LINT checks

```
%> vc_spyglass_import -sg_project <project-name> -goal <goal-name>
```

Once the above runs get completed then it compares the results:-

SG_INFO : Comparing Lint database..

SG_INFO : Comparison output file= '<dir-path>/lint.log'

SG_INFO : Diff Report= '<dir-path>/results.out'

<lint.log>

Template Database Generation Completed

SG moresimple report parsing completed (in 0s)

VC SG use model moresimple report parsing completed (in 0s)

Property and comparator registration completed

Report Comparison Completed (in 0s)

<results.out>

STRUCTURAL RULES

=====

Rule	Total (SG)	Total (VC)	Matched (SG)	Matched (VC)	Unmatched (SG)	Unmatched (VC)
W392	8	8	0	0	0	0

SG matched	SG unmatched	VC matched	VC unmatched	SG PASS%	VC PASS%
8	0	8	0	100%	100%

VC SpyGlass Project Import Kit Overview

LANGUAGE RULES

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Rule	Total (SG)	Total (VC)	Matched (SG)	Matched (VC)	Unmatched (SG)	Unmatched (VC)
W415a	257	257	257	257	0	0
W528	21	21	21	21	0	0
STARCO5-2.11.3.1	6	6	6	6	0	0
W240	22	22	22	22	0	0

SG matched	SG unmatched	VC matched	VC unmatched	SG PASS%	VC PASS%
306	0	306	0	100.00%	100.00%

How to Analyze Differences-

Comparison utility reports all the SpyGlass and VC SpyGlass only violations in the below files, please analyze these categories and try to find out the reason of differences.

SpyGlass Only:-

Work/sgUnmatches

VC SpyGlass Only:-

Work/vcUnmatches

How to invoke SpyGlass GUI?

```
%> vc_spyglass_import -sg_project <project-name> -goal <goal-name> -sg_gui
```

How to invoke VC SpyGlass LINT GUI?

```
%> vc_spyglass_import -sg_project <project-name> -goal <goal-name> -lint_gui
```

If you have moresimple.rpt file of SpyGlass run then provide the same with “-sg_moresimple” switch and in this case tool will not run the SpyGlass. It will directly invoke the VC SpyGlass in SGUM and gives the comparison data

```
%> vc_spyglass_import -sg_project <project-name> -goal <goal-name> -sg_moresimple <dir-path>/moresimple.rpt
```

4 Comparison of CDC Results

Provide the SpyGlass project file as input to “vc_spyglass_import”, it creates SpyGlass and VC SpyGlass setup files to perform desired CDC checks

```
%> vc_spyglass_import -sg_project <project-name> -goal <goal-name>
```

After completion of above run, it compares the CDC results

SG_INFO : Comparing CDC database..

SG_INFO : CDC summary= '<current-working-directory>/cdc.log'

SG_INFO : HTML Report= '<current-working-directory>/diff.html'

<cdc.log>

Parsing Files...

Done Parsing...

Comparing Setup...

Done Comparing Setup...

Comparing Rules...

Done Comparing Rules...

Dumping Results in HTML...

Process Complete !!

Section B#nos_bbox : MATCH=100%,EXCESS_IN_VC=0.00% (nothing_to_compare)

Section C#UDparam : MATCH=100%,EXCESS_IN_VC=0.00% (comparison_matched)

Section C#clock : MATCH=100%,EXCESS_IN_VC=0.00% (comparison_matched)

Section C#clock_domain : MATCH=100%,EXCESS_IN_VC=0.00% (comparison_matched)

Section C#reset : MATCH=100%,EXCESS_IN_VC=0.00% (comparison_matched)

Section C#set_case_analysis : MATCH=100%,EXCESS_IN_VC=0.00% (nothing_to_compare)

Section E#Crossings : MATCH=99.69%,EXCESS_IN_VC=0.21% (comparison_mismatched)

MATCH_ONLY_KEY_FIELDS=99.69%

Section E#Ac_unsync01, Ac_unsync02, Ac_sync01, Ac_sync02 Combined :

MATCH=88.66%,EXCESS_IN_VC=0.21% (comparison_mismatched)

MATCH_ONLY_KEY_FIELDS=99.69%

Section E#Ac_unsync01 : MATCH=100.00%,EXCESS_IN_VC=10.25% (comparison_matched)

Section E#Ac_unsync02 : MATCH=97.55%,EXCESS_IN_VC=21.41% (comparison_mismatched)
MATCH_ONLY_KEY_FIELDS=97.55%

Section E#Ac_sync01 : MATCH=90.39%,EXCESS_IN_VC=0.39% (comparison_mismatched)
MATCH_ONLY_KEY_FIELDS=90.39%

Section E#Ac_sync02 : MATCH=57.14%,EXCESS_IN_VC=5.66% (comparison_mismatched)
MATCH_ONLY_KEY_FIELDS=57.14%

<diff.html>

Section	Category	Run1 (Total Count)	Run2 (Total Count)	Previous	
				Run1 Only	Run2 Only
Section C: Setup Information	Number of User defined Parameter Values	14	14	0	0
Section C: Setup Information	Number of Clocks	5	5	0	0
Section C: Setup Information	Number of Clock Domains	5	5 (unique count), 6 (report count)	0	0
Section C: Setup Information	Number of Resets	5	5	0	0
Section C: Setup Information	Number of Propagated Clock signals	5	5	0	0
Section C: Setup Information	Number of Propagated Reset signals	5	5	0	0
Section D: Setup Errors	Setup_port01	45	45	0	0
Section E: CDC Analysis and Verification	Crossings	961	960	3	2
Section E: CDC Analysis and Verification	Ac_unsync01, Ac_unsync02, Ac_sync01, Ac_sync02 Combined	961	960	3	2
Section E: CDC Analysis and Verification	Ac_unsync01	232 (report-count) 219 (unique-count) (13 synchronized paths moved to Ac_sync01)	244	0	25
Section E: CDC Analysis and Verification	Ac_unsync02	361 (report-count) 286 (unique-count) (75 synchronized paths moved to Ac_sync02)	355	7	76
Section E: CDC Analysis and Verification	Ac_sync01	268 (report-count) 281 (unique-count) (13 synchronized paths moved from Ac_unsync01)	255	27	1
Section E: CDC Analysis and Verification	Ac_sync02	100 (report-count) 175 (unique-count) (75 synchronized paths moved from Ac_unsync02)	106	75	6

How to Analyze Differences-

Click on Run1 and Run2 only to check why crossings are coming in SpyGlass but not in VC SpyGlass and vice versa.

Crossings
In Run1 only [3 entries in 1 page(s)]
In Run2 only [2 entries in 1 page(s)]
Ac_unsync01, Ac_unsync02, Ac_sync01, Ac_sync02 Combined
In Run1 only [3 entries in 1 page(s)]
In Run2 only [2 entries in 1 page(s)]
Both in Run1 and Run2(with different attributes) [106 entries in 1 page(s)]
Ac_unsync01
In Run2 only [25 entries in 1 page(s)]
Ac_unsync02
In Run1 only [7 entries in 1 page(s)]
In Run2 only [76 entries in 1 page(s)]

How to invoke SpyGlass GUI?

```
%> vc_spyglass_import -sg_project <project-name> -goal <goal-name> -sg_gui
```

How to invoke VC SpyGlass CDC GUI?

```
%> vc_spyglass_import -sg_project <project-name> -goal <goal-name> -cdc_gui
```

If you have CDC-detailed-report.rpt file of SpyGlass run then you can provide the same with “-sg_detailed” and in this case tool will not run the SpyGlass. It will directly invoke the VC SpyGlass in SGUM and gives the comparison data.

```
%> vc_spyglass_import -sg_project <project-name> -goal <goal-name> -sg_detailed <dir-path>/CDC-detailed-report.rpt
```

5 Generate VCUM Setup Files

Use the “-generate_vc_setup” switch with “vc_spyglass_import” to generate the VCUM setup files for VC SpyGlass LINT/CDC/RDC.

```
%> vc_spyglass_import -sg_project <project-name> -goal <goal-name> -generate_vc_setup
```